

Literature Review – Review 2

Project: AI-Powered Social Media Crisis Detection and Early Warning System | Domain: Artificial Intelligence & Natural Language Processing | 10 Recent Research Papers

S.No.	Complete Authors	Research Paper Title	Journal / Publication	Methodology / Dataset	Key Findings	Research Gap / Limitation
1	Junaid Abdul Wahid; Mingliang Xu; Muhammad Ayoub; Xiaoheng Jiang; Shi Lei; Yufei Gao; Shabir Hussain; Yu Yang (2025)	AI-driven social media text analysis during crisis: A review for natural disasters and pandemics	Applied Soft Computing, 171, 112774	Machine Learning, Deep Learning, BERT, RoBERTa, Topic Modeling; review of 84 studies	Reviews AI/NLP methods for crisis-text analysis and highlights the growing role of deep learning and pretrained embeddings.	Need stronger real-time, explainable and low-resource/multilingual crisis-analysis systems.
2	Rabindra Lamsal; Maria Rodriguez Read; Shanika Karunasekera (2024)	CrisisTransformers: Pre-trained language models and sentence encoders for crisis-related social media texts	Knowledge-Based Systems, 296, 111916	Crisis-specific BERT/RoBERTa models and sentence encoders; 30+ crisis events; 18 public datasets	Crisis-specific transformer models outperformed strong baselines in classification; the best sentence encoder improved the state of the art by 17.43%.	Mainly improves text representation/classification; scope remains for integrated monitoring, trend analysis and early warning.
3	Abhishek Upadhyay; Yogesh Kumar Meena; Ganpat Singh Chauhan (2024)	SatCoBiLSTM: Self-attention based hybrid deep learning framework for crisis event detection in social media	Expert Systems with Applications, 249, 123604	Multi-scale CNN + BiLSTM + self-attention; three benchmark crisis datasets	Achieved F1 scores of 96%, 94% and 95% on three public datasets and improved over baselines.	Focuses on crisis-event detection; does not fully cover crisis growth, severity scoring and automated early-warning decisions.
4	Phyo Yi Win Myint; Siaw Ling Lo; Yuhao Zhang (2024)	Unveiling the dynamics of crisis events: Sentiment and emotion analysis via multi-task learning with attention mechanism and subject-based intent prediction	Information Processing & Management, 61(4), 103695	Transformer-based multi-task learning, attention, sentiment, emotion and intent prediction	Combines sentiment and emotion classification with attention and subject-based intent prediction for crisis tweets.	Signals are not fully combined into a real-time crisis-risk and alerting framework.
5	Nisha P. Shetty; Yash Bijalwan; Pranav Chaudhari; Jayashree Shetty; Balachandra Muniyal (2024/2025)	Disaster assessment from social media using multimodal deep learning	Multimedia Tools and Applications, 84, 18829–18854	Text + images; multimodal middle fusion; cross-modal attention and self-attention	Combining text and images improved disaster informativeness and disaster-type recognition; reported accuracies of 91.53% and 91.07%.	Multimodal analysis can be computationally demanding; lightweight real-time early-warning systems remain an opportunity.
6	Yiqiao Mao; Xiaoqiang Yan; Zirui Hu; Xuguang Zhang; Yangdong Ye; Hui Yu (2024)	Memory-aware continual learning with multi-modal social media streams for unsupervised disaster classification	Advanced Engineering Informatics, 62, 102654	Memory-aware continual learning, contrastive encoder and correlation memory bank; seven real-world datasets	Handles continuously arriving multimodal social-media information without repeatedly retraining from scratch.	Classification is the main goal; further work can connect continual monitoring to crisis evolution, severity and early warning.
7	Wang Gao; Changrui Zheng; Xun Zhu; Hongtao Deng; Yuwei Wang; Gang Hu (2024)	Knowledge-injected prompt learning for actionable information extraction from crisis-related tweets	Computers & Electrical Engineering, 118, 109398	Knowledge-injected prompt learning, word embeddings, few-shot and zero-shot learning	KAIEM extracts fine-grained actionable information with limited labelled data and improves precision, recall and F1 over baselines.	Actionable-information extraction is addressed, but not a complete crisis-risk scoring and alert-generation pipeline.
8	Wael Khallouli; Jiang Li; Jingwei Huang; Ghaith Rabadi; Samuel Kovacic (2025)	An integrated machine learning approach for identifying emergency rescue messages on social media during natural disasters	Social Network Analysis and Mining, 15, 50	Fine-tuned BERT + rule-based Regex; tweets from Hurricanes Harvey, Ian and Ida	Identifies emergency rescue requests and outperformed several ML/DL methods in AUC-PR across the three hurricane events.	Focuses on rescue-message identification rather than broader crisis monitoring, trend growth and general severity prediction.
9	Josip Katalinić; Ivan Dunđer (2025)	Neural Network-Based Sentiment Analysis and Anomaly Detection in Crisis-Related Tweets	Electronics, 14(11), 2273	Neural networks, deep-learning sentiment analysis and anomaly detection	Combines sentiment analysis with anomaly detection to identify unusual patterns that may signal important events or emerging hazards.	Could be extended by combining anomaly signals with topics, entities, emotion, trend growth and explainable crisis scoring.
10	Muhammad Mansoor; Zahoor Ur Rehman; Sitara Afzal; Joon-Min Gil; Jaechoon Jo; Yeo-Chan Yoon (2025)	NLP Framework for Disaster Prediction Using Social Media Analytics and Large Language Models	International Journal on Advanced Science, Engineering and Information Technology, 15(4), 1243–1249	Social-media analytics + fine-tuned RoBERTa/LLM approach	Uses social-media analytics and language models to improve disaster prediction and real-time crisis management.	Practical deployment still requires robust real-time monitoring, explainability and reliable alert/severity mechanisms.

Overall Identified Research Gap: Existing studies address individual tasks such as crisis-text classification, event detection, sentiment/emotion analysis, actionable-information extraction, anomaly detection and multimodal disaster assessment. However, there is scope for an integrated and explainable system that continuously monitors social-media activity, identifies rapidly growing issues, combines multiple NLP signals, estimates crisis severity and generates timely early warnings.